Question 4b.)

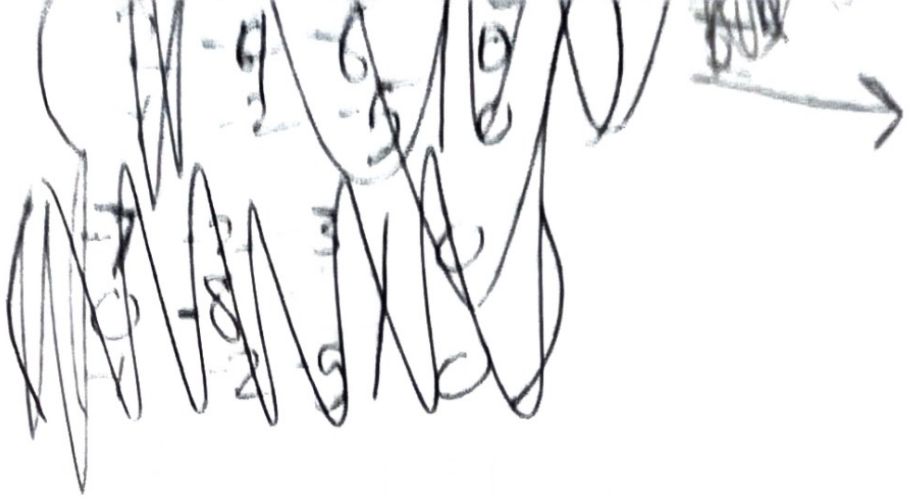
$\lambda_1 = 5$

$$A\vec{x} = \lambda\vec{x}$$

$$\begin{pmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = 5 \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$

$$-2x_1 + 2x_2 - 3x_3 = 5x_1$$
$$2x_1 + x_2 - 6x_3 = 5x_2$$
$$-1x_1 - 2x_2 = 5x_3$$

$$-7x_1 + 2x_2 - 3x_3 = 0$$
$$2x_1 - 4x_2 - 6x_3 = 0$$
$$-x_1 - 2x_2 - 5x_3 = 0$$



$$\left(\begin{array}{ccc|c} -7 & 2 & -3 & 0 \\ 2 & -4 & 6 & 0 \\ -1 & -2 & 5 & 0 \end{array}\right) \quad R1 \to R2$$

$$\left(\begin{array}{ccc|c} 2 & -4 & 6 & 0 \\ -7 & 2 & -3 & 0 \\ -1 & -2 & 5 & 0 \end{array}\right) \quad \frac{1}{2}R1 \to R1$$

$$\left(\begin{array}{ccc|c} 1 & -2 & -3 & 0 \\ -7 & 2 & -3 & 0 \\ -1 & -2 & -5 & 0 \end{array}\right) \quad R_1 + R_3 \to R3$$

$$\left(\begin{array}{ccc|c} 1 & -2 & -3 & 0 \\ -7 & 2 & -3 & 0 \\ 0 & -4 & -8 & 0 \end{array}\right) \quad 7R1 + R2 \to R2$$

$$\left(\begin{array}{ccc|c} 1 & -2 & -3 & 0 \\ 0 & -12 & -24 & 0 \\ 0 & -4 & -8 & 0 \end{array}\right)$$

$$\left(\begin{array}{ccc|c} 1 & -2 & -3 & 0 \\ 0 & -12 & -24 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right) \xrightarrow{-3R3+R2 \to R2}$$

X_3 is an ambiguous solution
"free variable"

$X_3 = 1$

$-12X_2 - 24X_3 = 0$

$X_2 = \frac{24X_3}{-12} = \frac{24(1)}{-12}$

$X_2 = -2$

$X_1 + 4 - 3 = 0$

$X_1 + 1 = 0$

$X_1 = -1$

J

$X = \begin{pmatrix} -1 \\ -2 \\ +1 \end{pmatrix}$

for $\lambda = 5$